

Decent Coaching Classes
IX CBSE
Academic Conquest - III
Answer Key — Chapter 8: Journey Inside the Atom

SECTION - A

1. (b) J.J. Thomson — he studied cathode rays passing through gases at low pressure and identified them as streams of negatively charged particles (electrons).
2. (b) Planetary model — Rutherford's model describes electrons revolving around a central nucleus, similar to planets orbiting the Sun.
3. (c) Neutron — discovered by James Chadwick in 1932; it has a mass nearly equal to that of a proton but carries no electrical charge.
4. (c) 18 — using the formula $2n^2$, for M-shell $n = 3$, so $2 \times 3^2 = 18$.
5. (b) Isotopes — by definition, isotopes are atoms of the same element (same atomic number) with different mass numbers due to differing neutron numbers.
6. (a) Both A and R are true, and R is the correct explanation of A — most alpha particles passed straight through the gold foil because most of the volume of an atom is empty space, which is precisely why so few particles were deflected or bounced back.

SECTION - B

7. Atomic number (Z) is the number of protons present in the nucleus of an atom of an element. Mass number (A) is the total number of protons and neutrons (nucleons) present in the nucleus of an atom.
8. Any two of the following: (i) Electrons revolve around the nucleus only in fixed circular paths called stationary states/orbits/shells, each having a definite amount of energy. (ii) While moving in a fixed shell, an electron does not lose energy. (iii) An electron can move from one shell to another by absorbing or releasing energy exactly equal to the difference between the energies of the two shells.
9. Chlorine (Z = 17): electronic configuration is 2, 8, 7 (K = 2, L = 8, M = 7). It has 7 valence electrons (electrons in the outermost M-shell).
10. Valency is the number of electrons that an atom gains, loses, or shares to complete its octet (8 electrons) in the outermost shell and attain a stable configuration. Sodium's electronic configuration (2, 8, 1) has 1 valence electron; since this is fewer than 4, sodium loses this one electron to attain a stable octet. Hence, the valency of sodium is 1.

SECTION - C

11. In 1911, Geiger and Marsden (working under Rutherford) directed a narrow beam of alpha particles at an extremely thin gold foil and observed the particles' paths using a fluorescent screen surrounding the foil. Any two observations: (i) Most of the alpha particles passed straight through the gold foil without any deflection. (ii) A small number of alpha particles were deflected through small or large angles. (iii) A very small number of alpha particles (about 1 in 20,000) bounced back almost along their original path. Conclusion: the positive charge and most of the mass of the atom are concentrated in a very small region called the nucleus, while the rest of the atom is mostly empty space.
12. According to the established law of physics, a charged particle moving in a circular path is constantly accelerating (since its direction keeps changing) and hence continuously loses energy in the form of radiation. In Rutherford's model, electrons revolve around the nucleus in circular paths, so they should continuously lose energy, spiral inward, and eventually fall into the nucleus. This would make the atom unstable and cause it to collapse. However, in reality, atoms (other than radioactive ones) are known to be stable. Therefore, Rutherford's model failed to explain the stability of the atom.

OR

- Bohr proposed that electrons revolve around the nucleus only in certain fixed circular orbits called stationary states or shells, each with a definite, constant amount of energy. As long as an electron remains in a particular stationary orbit, it does not lose or gain energy — unlike what classical physics would predict for an accelerating charge. An electron can move to another orbit only by absorbing or releasing energy exactly equal to the difference in energy between the two orbits. Since the electron's energy remains fixed while in a stable orbit, it does not spiral into the nucleus, and the atom remains stable. This successfully addressed the shortcoming of Rutherford's model.
13. Oxygen (2, 6) has 6 valence electrons, which is more than 4. To complete its octet, it needs 2 more electrons, so it tends to gain 2 electrons. Hence, the valency of oxygen is 2. Carbon (2, 4) has exactly 4 valence electrons — neither less than 4 nor

more than 4 — so it cannot easily lose or gain electrons. Instead, it shares 4 electrons with other atoms to complete its octet. Hence, the valency of carbon is 4.

14. Average atomic mass = (mass of isotope 1 $\times$ % abundance/100) + (mass of isotope 2 $\times$ % abundance/100)

$$\text{Average atomic mass} = (79 \times 49.7/100) + (81 \times 50.3/100)$$

$$= 39.263 + 40.743$$

$$= 80.006 \text{ u} \approx 80.0 \text{ u}$$

Hence, the average atomic mass of bromine is approximately 80.0 u.

SECTION - D

15. Solution:

	X	Y	Z
Protons	18	17	17
Neutrons	19	18	20
Mass number	37	35	37

(i) Mass number = number of protons + number of neutrons.

$$\text{X: } 18 + 19 = 37 \quad \text{Y: } 17 + 18 = 35 \quad \text{Z: } 17 + 20 = 37$$

(ii) Y and Z have the same number of protons (17), i.e., the same atomic number, but different mass numbers (35 and 37). Atoms of the same element with the same atomic number but different mass numbers are called isotopes. So, Y and Z are isotopes of the same element (chlorine).

(iii) Z and X have different atomic numbers (17 and 18) but the same mass number (37). Atoms of different elements having the same mass number but different atomic numbers are called isobars. So, Z and X are isobars.

Total: 30 Marks